Vector Spaces and Subspaces

T.T.

Q1

(i) The set S_1 is a subspace. It contains the zero vector, and if $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ belong to S_1, then for any $c, d \in \mathbb{R}$,

$$c(x_1 + y_1 + z_1) + d(x_2 + y_2 + z_2) = 0.$$

Thus $cu + dv \in S_1$.

The set S_2 is not a subspace because it does not contain the zero vector.

The set S_3 is not a subspace. For example,

$$u = (1, 0, 1), \qquad v = (0, 1, 1)$$

both belong to S_3, but $u + v = (1, 1, 2)$ does not.

The set S_4 is a subspace by the definition of span. Any linear combination of vectors in S_4 is again a linear combination of $(1, 4, 0)$ and $(2, 2, 2)$.

The set S_5 is not a subspace. The vector $(0, 1, 2)$ belongs to S_5, but its negative $(0, -1, -2)$ does not satisfy $0 \le -1 \le -2$.

(ii) Because S and T are subspaces, both contain the zero vector. Hence $0 \in S \cap T$. Let $u, v \in S \cap T$ and let $c, d \in \mathbb{R}$. Since $u, v \in S$ and S is a subspace,

$$cu + dv \in S.$$

The same argument gives $cu + dv \in T$. Therefore $cu + dv \in S \cap T$, and $S \cap T$ is a subspace of V.

Q2

(i) Every vector in the column space of $[A\ b]$ has the form

$$Ax + tb, \qquad x \in \mathbb{R}^n, \quad t \in \mathbb{R}.$$

Suppose first that $b \in \mathcal{C}(A)$. Then $b = Ax_0$ for some $x_0 \in \mathbb{R}^n$, and

$$Ax + tb = A(x + tx_0) \in \mathcal{C}(A).$$

Thus $\mathcal{C}([A\ b]) \subseteq \mathcal{C}(A)$. The reverse inclusion follows by taking $t = 0$, so the two column spaces are equal.

Conversely, suppose $\mathcal{C}([A\ b]) = \mathcal{C}(A)$. The appended column b belongs to $\mathcal{C}([A\ b])$, and therefore $b \in \mathcal{C}(A)$. Finally, by the definition of the column space,

$$b \in \mathcal{C}(A) \quad \Longleftrightarrow \quad b = Ax \text{ for some } x \in \mathbb{R}^n.$$

This is exactly the solvability of $Ax = b$.

(ii) Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Its column space is the x-axis in $\mathbb{R}^2$. Appending

$$b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

enlarges the column space to all of $\mathbb{R}^2$. By contrast, appending

$$c = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

does not enlarge it because c already belongs to $\mathcal{C}(A)$.

Q3

(i) With

$$A = \begin{bmatrix} 1 & -3 & -1 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

the equation is $A\mathbf{x} = 0$. The variable x is a pivot variable, while y and z are free. Solving for the pivot variable gives

$$x = 3y + z.$$

Therefore

$$\mathbf{x} = y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

A basis for the nullspace is

$$\left\{ \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

(ii) A convenient particular solution is

$$\mathbf{x}_p = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}.$$

The complete solution is

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.$$

This solution set is not a subspace because it does not contain the zero vector. It is an affine plane obtained by translating $\mathcal{N}(A)$ by $\mathbf{x}_p$.

Q4

(i) Elimination on the augmented matrix begins with

$$\begin{bmatrix} 1 & 2 & -2 & b_1 \\ 2 & 5 & -4 & b_2 \\ 4 & 9 & -8 & b_3 \end{bmatrix}.$$

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 4R_1$:

$$\begin{bmatrix} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 1 & 0 & b_3 - 4b_1 \end{bmatrix}.$$

Then $R_3 \leftarrow R_3 - R_2$ gives

$$\begin{bmatrix} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - b_2 - 2b_1 \end{bmatrix}.$$

The system is solvable if and only if

$$b_3 - b_2 - 2b_1 = 0.$$

(ii) Assume that $b_3 = b_2 + 2b_1$. The second equation gives

$$y = b_2 - 2b_1.$$

Choose the free variable $z = t$. The first equation then gives

$$x = 5b_1 - 2b_2 + 2t.$$

Hence the complete solution is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5b_1 - 2b_2 \\ b_2 - 2b_1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

The first vector is one particular solution, and the second term contains all homogeneous solutions.

(iii) The echelon form has two pivots, so the coefficient matrix has rank 2. Its nullspace is one-dimensional, with basis

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Q5

Place u_1, u_2, u_3 into the columns of a matrix:

$$U = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}.$$

Apply $R_2 \leftarrow R_2 - 3R_1$ and $R_3 \leftarrow R_3 - 2R_1$, then exchange rows 2 and 3:

$$U \longrightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & -5 & -7 \end{bmatrix}.$$

Finally, $R_3 \leftarrow R_3 - 5R_2$ gives

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 18 \end{bmatrix}.$$

There are three pivots, so the only solution of

$$c_1u_1 + c_2u_2 + c_3u_3 = 0$$

is $c_1 = c_2 = c_3 = 0$. Thus u_1, u_2, u_3 are linearly independent. Three independent vectors in $\mathbb{R}^3$ form a basis for $\mathbb{R}^3$.

For the second triple,

$$v_1 + v_2 + v_3 = (0, 0, 0).$$

This is a nontrivial linear relation, so v_1, v_2, v_3 are dependent and do not form a basis for $\mathbb{R}^3$.

Q6

(i) Every polynomial in $\mathcal{P}_3$ has a unique representation

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

Therefore

$$\{1, x, x^2, x^3\}$$

is a basis for $\mathcal{P}_3$, and $\dim \mathcal{P}_3 = 4$.

(ii) The zero polynomial belongs to S. If $p, q \in S$ and $c, d \in \mathbb{R}$, then

$$(cp + dq)(1) = cp(1) + dq(1) = 0.$$

Thus $cp + dq \in S$, and S is a subspace.

The condition $p(1) = 0$ gives

$$a_0 + a_1 + a_2 + a_3 = 0,$$

so $a_0 = -a_1 - a_2 - a_3$. Consequently,

$$p(x) = a_1(x - 1) + a_2(x^2 - 1) + a_3(x^3 - 1).$$

The three polynomials $x - 1$, $x^2 - 1$, and $x^3 - 1$ are linearly independent because their leading nonzero terms have different degrees. Hence

$$\{x - 1, x^2 - 1, x^3 - 1\}$$

is a basis for S, and $\dim S = 3$.

Q7

(i) The second row of A is twice the first, and every column is a multiple of $(1,2)$. Therefore $\text{rank}(A) = 1$. A basis for the column space is

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}, \qquad \dim \mathcal{C}(A) = 1,$$

and a basis for the row space is

$$\{(1,2,4)\}, \qquad \dim \mathcal{R}(A) = 1.$$

The equation $Ax = 0$ reduces to

$$x_1 + 2x_2 + 4x_3 = 0.$$

Thus

$$x = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix},$$

so a basis for the nullspace is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} \right\}, \qquad \dim \mathcal{N}(A) = 2.$$

Finally, $A^T y = 0$ reduces to $y_1 + 2y_2 = 0$. A basis for the left nullspace is

$$\left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}, \qquad \dim \mathcal{N}(A^T) = 1.$$

(ii) The first two columns of B are independent because

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = 0$$

implies $c_1 + 2c_2 = 0$ and $2c_1 + 5c_2 = 0$, hence $c_1 = c_2 = 0$. Therefore $\text{rank}(B) = 2$. A basis for the column space is

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}, \qquad \dim \mathcal{C}(B) = 2.$$

The two rows are also independent, so a basis for the row space is

$$\{(1,2,4),(2,5,8)\}, \qquad \dim \mathcal{R}(B) = 2.$$

Solving $Bx = 0$ gives

$$x_2 = 0, \qquad x_1 = -4x_3.$$

Therefore a basis for the nullspace is

$$\left\{ \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} \right\}, \qquad \dim \mathcal{N}(B) = 1.$$

Since the columns span $\mathbb{R}^2$, the left nullspace contains only the zero vector. Its basis is empty and

$$\dim \mathcal{N}(B^T) = 0.$$

(iii) For A, the dimensions satisfy

$$\dim \mathcal{R}(A) + \dim \mathcal{N}(A) = 1 + 2 = 3$$

and

$$\dim \mathcal{C}(A) + \dim \mathcal{N}(A^T) = 1 + 1 = 2.$$

The row-space basis vector is orthogonal to both nullspace basis vectors:

$$(1, 2, 4) \cdot (-2, 1, 0) = 0, \qquad (1, 2, 4) \cdot (-4, 0, 1) = 0.$$

The column-space and left-nullspace basis vectors are also orthogonal:

$$(1, 2) \cdot (-2, 1) = 0.$$

For B, the dimensions satisfy

$$\dim \mathcal{R}(B) + \dim \mathcal{N}(B) = 2 + 1 = 3$$

and

$$\dim \mathcal{C}(B) + \dim \mathcal{N}(B^T) = 2 + 0 = 2.$$

The nullspace basis vector is orthogonal to both rows:

$$(1, 2, 4) \cdot (-4, 0, 1) = 0, \qquad (2, 5, 8) \cdot (-4, 0, 1) = 0.$$

The left nullspace is the zero space, which is orthogonal to every vector in $\mathcal{C}(B)$.

Q8

(i) Write the columns of B as $b_1, \ldots, b_p$. Then

$$AB = \begin{bmatrix} Ab_1 & \cdots & Ab_p \end{bmatrix}.$$

Every column Ab_j is a linear combination of the columns of A. Therefore

$$\mathcal{C}(AB) \subseteq \mathcal{C}(A).$$

Taking dimensions gives

$$\operatorname{rank}(AB) \leq \operatorname{rank}(A).$$

(ii) If $x \in \mathcal{N}(B)$, then $Bx = 0$, and consequently

$$ABx = A0 = 0.$$

Thus $x \in \mathcal{N}(AB)$, proving that

$$\mathcal{N}(B) \subseteq \mathcal{N}(AB).$$

Both nullspaces are subspaces of $\mathbb{R}^p$, so

$$\dim \mathcal{N}(B) \leq \dim \mathcal{N}(AB).$$

By rank-nullity,

$$p - \operatorname{rank}(B) \leq p - \operatorname{rank}(AB),$$

and hence

$$\text{rank}(AB) \le \text{rank}(B).$$

(iii) For every $x \in \mathbb{R}^n$,

$$(C + D)x = Cx + Dx.$$

Therefore every vector in $\mathcal{C}(C + D)$ belongs to the sum of subspaces

$$\mathcal{C}(C) + \mathcal{C}(D) = \{u + v : u \in \mathcal{C}(C),\ v \in \mathcal{C}(D)\}.$$

If bases for $\mathcal{C}(C)$ and $\mathcal{C}(D)$ contain r_C and r_D vectors, their union spans $\mathcal{C}(C) + \mathcal{C}(D)$. Consequently,

$$\begin{aligned}
\text{rank}(C + D) &= \dim \mathcal{C}(C + D) \\
&\le \dim\big(\mathcal{C}(C) + \mathcal{C}(D)\big) \\
&\le r_C + r_D \\
&= \text{rank}(C) + \text{rank}(D).
\end{aligned}$$

(iv) Take

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Both A and B have rank 1, while $AB = 0$ has rank 0. Hence

$$\text{rank}(AB) = 0 < \text{rank}(A) = 1$$

and

$$\text{rank}(AB) = 0 < \text{rank}(B) = 1.$$